

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building and scaling large-scale platforms, distributed systems, and AI-powered products. Experience spans product development, platform engineering, architecture modernization, and applied AI, with systems operating at 100K+ TPS, platforms adopted across 60+ services, and products delivering \$28M+ in measured annualized impact.

EXPERIENCE**Senior Software Engineer | Microsoft | Redmond, WA****Sep 2025 to Aug 2026**

- Spearheaded Microsoft's flagship enterprise AI Agent Risk Detection platform as lead architect, orchestrating 90-day activity reconstruction across 40+ global environments to unlock instant day-one risk governance for 237,000+ AI agents across 13,000 enterprise customers.
- Directed principal-level architecture across Purview IRM, Entra, and DLP for Microsoft's flagship AI Adaptive Protection pipeline; authored the end-to-end technical spec and shipped core conditional access enforcement powering Microsoft 365 E7.
- Pioneered hyper-scale distributed ingestion infrastructure with zero-latency fault isolation, partitioning agent and human traffic across 28 worldwide deployments processing 150K+ critical security signals/day at 99.999% SLA.
- Championed sovereign-cloud infrastructure expansion across US GCC, GCCH, and DoD high-security enclaves, orchestrating 53 Azure resource groups and 51 deployments with automated disaster recovery and real-time Kusto telemetry.
- Orchestrated unified investigation graphs across Entra, Defender, Sentinel, and Graph, delivering unified threat intelligence that boosted customer security operation engagement by 28%.
- Architected autonomous AI-agent synthetic validation and end-to-end test harnesses using Playwright, simulating advanced attack vectors and policy violations across dynamic enterprise environments.
- Pioneered closed-loop AI debugging and self-healing pipelines across 3 major codebases, integrating telemetry pattern recognition with Playwright to automatically reproduce issues, generate code fixes, and submit PRs.

Software Engineer 2 | Amazon | Seattle, WA**Aug 2019 – Aug 2025**

- Built a 0→1 developer platform for 60+ microservices**, creating a one-click CDK-based workflow that standardized infrastructure, authentication, observability, dependencies, CI/CD, and deployment into a production-ready service; cut setup from 2 weeks to under 1 hour, consolidated 9 pipelines, and saved 40–50 engineering weeks across ~40 engineers and 3 teams.
- Led a monolith-to-microservices transformation at 100K+ TPS**, building and launching 15+ services and establishing shared REST/gRPC orchestration supporting 10M+ daily transactions while maintaining behavioral parity throughout migration.
- Designed and owned the service powering ML-predicted delivery ranges**, integrating a quantile-regression model and coordinating changes across inventory planning and fulfillment optimization systems to replace single-date promises; delivered \$28M in annualized impact.
- Replaced static delivery-risk rules with ML-driven decisioning**, using XGBoost on SageMaker to identify at-risk orders and trigger recovery, customer notifications, and alternate-seller recommendations, reducing cancellations and contributing to a 7% increase in sales.
- Launched inventory decisioning systems processing 200K+ events/day**, launching Just-In-Stock and Restock Alerting with 30-day order replay and real-time inventory signals; increased restocked-item sales 13% and early deliveries 16%.
- Architected built a RAG-based product recommendation system for out-of-stock items**, to surface relevant alternatives grounded in real-time inventory and delivery constraints, keeping availability and fulfillment validation deterministic. Increased eligible product sales by 23%.
- Led design and cross-functional adoption of an LLM-powered localization platform**, using Bedrock with vocabulary-constrained generation and deterministic validation to personalize 37 email templates across 28 locales, with pre-approved template fallbacks on validation failure to address hallucination and compliance risks.
- Owned the annual operational improvement roadmap across 60+ services**, driving planning and execution across 15+ engineers and reducing Sev2 incidents 40% in two months; led readiness across 90 services for Prime Day and Black Friday with zero major incidents.
- Turned synchronized delivery for large Amazon Business orders into a first-class fulfillment capability**, driving changes across planning, inventory, and fulfillment systems to honor coordinated delivery dates; replaced manual engineering workflows with self-service tooling, eliminated 100–300 engineering hours/year, and improved GCCP 11%.
- Standardized observability and container infrastructure across 60+ services**, introducing shared logging and Docker libraries that cut CloudWatch spend \$250K+/month, container size 83%, and build time 50%, with container standards adopted across teams.

Software Engineering Intern | Liquiron | San Jose, CA**Dec 2018 to Jan 2019**

- Migrated 2 backend services to serverless**, modernizing authentication with OAuth 2.0/Passport.js and Firebase.
- Built the API layer for a single-page application**, delivering 4 REST APIs, 10 frontend capabilities and reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India**Sep 2016 to Jul 2017**

- Built a real-time CNN classification system for images and video**, achieving 87% accuracy across 7 sentiment categories.
- Modernized legacy web applications into a single-page architecture**, improving response time 25% and unreliable-network performance 9%; introduced an engineering feedback system adopted by 6 teams.

Software Engineering Intern | Soft Corner | Pune, India**Jun 2015 to Jun 2016**

- Improved F2FS garbage-collection efficiency 15% through Linux kernel-level performance engineering**, instrumenting filesystem behavior and benchmarking cost-benefit and greedy cleaning policies to optimize latency.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. Sep 2017 to Jun 2019
- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. Jun 2012 to Jun 2016

FEATURED PROJECTS

- LeetDesign**: Executable system-design simulator for modeling scalability, fault tolerance, workloads, and architectural trade-offs.
- FinanceOS**: AI Financial Intelligence Platform: Designed a hybrid deterministic + LLM architecture that computes financial facts through deterministic pipelines and uses LLM reasoning only for explanation and recommendations, keeping probabilistic models outside correctness-critical calculations.

SKILLS

- Languages**: Java • C# • Python • C++ • C • TypeScript • JavaScript • SQL
- Architecture & Systems**: Distributed Systems • System Design • Microservices • Event-Driven Architecture • REST APIs • gRPC • Concurrency • Fault Tolerance • High Availability • Linux
- Cloud & Platform**: AWS • Azure • Platform Engineering • Developer Platforms • Infrastructure as Code (IaC) • CI/CD • Docker • ECS • Lambda • Step Functions • DynamoDB • Cosmos DB • Observability • SLOs
- AI/ML & Security**: Generative AI • LLMs • RAG • AI Agents • AI Infrastructure • AI Governance • Machine Learning • NLP • XGBoost • Bedrock • SageMaker • AI Security • DLP • Entra • Purview • Playwright